

Renal Tubular Dysgenesis, ACE Type (RTD-ACE) — Comprehensive Disease Report

Disease: Renal Tubular Dysgenesis, ACE type (RTD-ACE) **MONDO ID:** MONDO:0700337 · **OMIM:** 267430 (Renal Tubular Dysgenesis) · **Orphanet:** ORPHA:97369 · **Category:** Mendelian (autosomal recessive) **Causal gene:** *ACE* (angiotensin I-converting enzyme; HGNC:2707; gene OMIM 106180; chromosome 17q23.3)

Summary

Renal Tubular Dysgenesis of the ACE type (RTD-ACE) is a rare, autosomal recessive developmental disorder of the kidney caused by biallelic loss-of-function (LoF) variants in *ACE*, the gene encoding angiotensin I-converting enzyme. It is one subtype within a broader genetic RTD spectrum produced by inactivating mutations in any of the four core renin-angiotensin system (RAS) genes — *AGT* (angiotensinogen), *REN* (renin), *ACE*, and *AGTR1* (angiotensin II receptor type 1). The defining lesion is **fetal angiotensin II deficiency**: without functional ACE, angiotensin I cannot be converted to the active octapeptide angiotensin II, the fetal RAS is inactive, renal perfusion pressure collapses, proximal tubules fail to differentiate, and the fetus becomes anuric. The clinical hallmark is a combination of persistent fetal anuria → oligohydramnios/anhydramnios → Potter sequence with pulmonary hypoplasia, together with refractory neonatal hypotension and defective skull ossification (large fontanelles, wide sutures). Perinatal mortality is very high.

Across genetically solved RTD families, *ACE* is the single most common cause, accounting for roughly two-thirds (≈65%) of families (Gribouval et al. 2012, [PMID: 22095942](#)). Critically, disease severity is similar regardless of which RAS gene is mutated, which points strongly to a **single unifying mechanism — loss of fetal angiotensin II signaling** — rather than any ACE-specific effect. This human genetic convergence is mirrored in mouse genetics: knockouts of *Ace*, *Agt*, and the combined *Agtr1a/Agtr1b* receptors all produce a nearly identical renal phenotype of vascular thickening, medullary atrophy/tubular lesions, hypotension, and perinatal lethality.

Although RTD-ACE was historically considered uniformly lethal, the clinical spectrum is expanding. With aggressive neonatal blood-pressure support, mechanical ventilation, and early dialysis, survivors are now reported living from years into adulthood with variable chronic kidney disease (CKD). A distinct **atypical presentation** of biallelic *ACE* LoF is also recognized: progressive childhood CKD with polyuria–polydipsia and anemia that is disproportionate to the degree of renal failure. This disproportionate anemia is mechanistically explained by the loss of angiotensin II's direct pro-erythropoietic action (both as an erythropoietin secretagogue and as a direct AT1-mediated growth factor for erythroid progenitors) — the same angiotensin II deficiency that drives the renal phenotype.

Key Findings

Finding 1 — RTD-ACE is caused by biallelic loss-of-function variants in *ACE*, within a RAS-gene RTD spectrum

RTD is genetically heterogeneous but mechanistically unified. The landmark study by Gribouval et al. 2005 ([PMID: 16116425](#)) identified homozygous or compound heterozygous mutations in the genes encoding renin, angiotensinogen, angiotensin-converting enzyme, or angiotensin II receptor type 1 in 11 individuals from 9 families with autosomal recessive RTD, establishing that *ACE* is one of four RAS genes whose biallelic inactivation causes the disease. The authors wrote that affected individuals *"had homozygous or compound heterozygous mutations in the genes encoding renin, angiotensinogen, angiotensin converting enzyme or angiotensin II receptor type 1."*

Multiple subsequent *ACE*-specific case reports and series confirmed biallelic LoF *ACE* variants — spanning missense, frameshift, nonsense, and splice-site classes — as the most common genetic cause of RTD. For example, novel compound heterozygous *ACE* variants were reported in a family with recurrent anhydramnios ([PMID: 32329243](#)), a homozygous novel 3' splice-site variant (NM_000789.3:c.2642-1G>A) was found in a fetus with RTD ([PMID: 30058238](#)), and *ACE* variants were associated with RTD plus a novel microcolon association ([PMID: 30071301](#)). The progressive-CKD review by Fila et al. 2020 ([PMID: 32198635](#)) states plainly: *"Bi-allelic loss of function variations in genes encoding proteins of the renin-angiotensin system (AGT, ACE, REN, AGTR1) are associated with autosomal recessive renal tubular dysgenesis."*

Variant / molecular detail: - **Gene:** *ACE* (HGNC:2707), 17q23.3; reference transcript NM_000789.3. - **Variant types observed:** missense (e.g., p.Ser346Trp, p.Asn489Asp), frameshift (p.S486Ffs29), *nonsense, and canonical splice-site (c.2642-1G>A)*. - **Functional consequence:** *loss of function — loss of ACE catalytic activity → no angiotensin II generation. In vitro/in silico analyses show such variants disrupt ACE protein structure (PMID: 32329243)*. - **Classification:*** reported variants are generally pathogenic/likely pathogenic under ACMG/AMP criteria given the LoF mechanism, recessive segregation, and absence/rarity in population databases.

Finding 2 — Mechanism: fetal RAS inactivity causes low renal perfusion, proximal-tubule maldevelopment, anuria, and Potter sequence

The histopathology of RTD directly demonstrates the developmental lesion. Lacoste et al. 2006 (PMID: 16790508) found that *"Severe defects in proximal tubules were observed in all fetuses from 18 gestational weeks onward,"* establishing the timing of the proximal-tubule differentiation failure. The same study documented a strikingly increased renal renin signal — *"Renal renin expression was strikingly increased in 19 of 24 patients studied"* — reflecting compensatory upregulation of renin in the face of absent downstream angiotensin II signaling (a feedback hallmark of RAS failure). Thickened renal arterial vasculature (from arcuate to afferent arteries) is also characteristic, and ACE/angiotensinogen protein is absent or reduced in the proximal tubule.

Gribouval et al. 2005 (PMID: 16116425) articulated the core causal chain: *"renal lesions and early anuria result from chronic low perfusion pressure of the fetal kidney, a consequence of renin-angiotensin system inactivity."* An earlier immunohistochemical study (PMID: 7994002) had already noted large accumulations of renin in preglomerular arterioles and glomerular structures, consistent with faulty feedback and reduced glomerular perfusion.

Clinically, this produces a stereotyped cascade: persistent fetal anuria → oligohydramnios/anhydramnios → pulmonary hypoplasia and limb/facial deformation (Potter sequence); refractory neonatal hypotension unresponsive to vasopressors; and defective skull ossification with large fontanelles and wide cranial sutures (reported consistently across cases, e.g., PMID: 32808512, PMID: 38649831).

Finding 3 — High perinatal mortality but an expanding survivor spectrum, including progressive CKD

Classically, RTD causes near-universal perinatal death from the combination of pulmonary hypoplasia and anuric renal failure — in the Lacoste series all 29 cases resulted in perinatal death. However, the modern clinical picture is shifting. Schreiber et al. 2021 (PMID: 34957720) reported a cohort of survivors in whom *"the patients are 5-20 years old with variable stages of chronic kidney disease,"* with 4 of 5 carrying *ACE* mutations. Numerous recent single-case reports describe neonates surviving the neonatal period with early peritoneal dialysis and intensive supportive management (PMID: 42831903, PMID: 40930738, PMID: 40709124, PMID: 35848000).

Biallelic *ACE* LoF can also present **atypically** as progressive childhood CKD rather than lethal neonatal disease. Fila et al. 2020 (PMID: 32198635) reported that *"Bi-allelic loss of function mutation of ACE can have atypical and sometimes late presentation with chronic renal failure, anemia (out of proportion with the level of renal failure), and polyuro-polydipsia,"* broadening the recognized phenotypic range and emphasizing disproportionate anemia and a urinary concentrating defect as key atypical features. Survival with managed tubular and glomerular dysfunction has even been documented to adulthood in a related *AGTR1*-LoF patient (28 years old, GFR ~30 mL/min; PMID: 33768328).

Finding 4 — Mouse *RAS*-gene knockouts recapitulate the RTD renal vascular/tubular pathology and hypotension

The mouse genetics provide convergent, cross-species confirmation of the mechanism. *Ace*-null mice *"have low systolic blood pressures and defects in renal development and function"* (PMID: 18406804). Combined *AT1* receptor knockouts (*Agtr1a*^{-/-}*Agtr1b*^{-/-}) show vascular thickening within the kidney and atrophy of the inner renal medulla, a phenotype described as *"virtually identical to that seen in angiotensinogen-deficient (Agt^{-/-}) and angiotensin-converting enzyme-deficient (Ace^{-/-}) mice that are unable to synthesize angiotensin II"* (PMID: 9860997). *Agt*^{-/-} mice die shortly after birth from renal dysfunction with severe renal vascular and tubular lesions (PMID: 11096065). Finally, mice with a catalytically inactivated *ACE* C-domain cannot concentrate urine effectively after dehydration — *"they are not able to concentrate urine after dehydration as effectively as wild-type mice"* (PMID: 18158355) — modeling the tubular concentrating defect seen in human RTD survivors.

Finding 5 — ACE is the predominant molecular cause (~65% of families), with gene-independent disease severity

The definitive genotype–phenotype survey, Gribouval et al. 2012 ([PMID: 22095942](#)), reviewed 54 distinct RAS-gene mutations across 48 unrelated families with autosomal recessive RTD. ACE mutations were the most frequent, found in **two-thirds of families (64.6%)**, and — crucially — *"The severity of the clinical course was similar whatever the mutated gene."* This gene-independence is the strongest human-genetic argument that a single downstream endpoint (angiotensin II deficiency / renal hypoperfusion) drives the disease. The same review generalizes the mechanism to acquired forms: *"Renal hypoperfusion, whether genetic or secondary to a variety of diseases, precludes the normal development/ differentiation of proximal tubules."* Developmental studies (Yosypiv, [PMID: 24061643](#); [PMID: 21359618](#)) add that loss of angiotensin II impairs ureteric-bud branching morphogenesis, causing medullary/papillary hypoplasia and the urinary concentrating defect.

Finding 6 — Disproportionate anemia in ACE-RTD survivors reflects loss of angiotensin II-driven erythropoiesis

The anemia-out-of-proportion feature of ACE-LoF survivors ([PMID: 32198635](#)) has a clear mechanistic basis. Angiotensin II is a physiological regulator of erythropoiesis, acting as both an erythropoietin secretagogue and a direct AT1-mediated growth factor for erythroid progenitors — Vlahakos et al. 2010 ([PMID: 20400218](#)) state that *"angiotensin II (Ang II) is a physiologically important regulator of erythropoiesis."* Genetic proof comes from Kato/Fukamizu 2015 ([PMID: 26107632](#)): *"knockout mice that lack angiotensin II, including angiotensinogen and renin knockout mice, exhibit anemia. The anemia of angiotensinogen knockout mice was rescued by angiotensin II infusion"* — and the rescue was re-blocked by an AT1 antagonist, while AT1a/AT1b double-knockout mice reproduced the anemia. Thus the same angiotensin II deficiency that drives renal maldevelopment independently impairs red-cell production, explaining why anemia in ACE-RTD is worse than renal EPO deficiency alone would predict.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. Biallelic loss-of-function variant in ACE (missense/frameshift/nonsense/splice)
| results in
2. Absent/deficient angiotensin I-converting enzyme catalytic activity
| results in
3. Failure to convert angiotensin I → angiotensin II (fetal RAS inactive)
| leads to
4. Loss of AT1-receptor signaling + chronically low fetal renal perfusion pressure
| branches into three downstream arms:
 - ARM A (renal development – demonstrated histologically):
 - 5a. Impaired proximal-tubule differentiation (from ~18 wk gestation)
+ impaired ureteric-bud branching morphogenesis (inferred from mouse/dev studies)
-> medullary/papillary hypoplasia, thickened preglomerular vasculature
-> fetal anuria → oligohydramnios/anhydramnios
-> Potter sequence (pulmonary hypoplasia, limb/facial deformation)
-> neonatal respiratory failure <- major cause of perinatal death
 - ARM B (hemodynamic):
 - 5b. Loss of angiotensin II vasopressor tone
-> refractory neonatal hypotension (vasopressor-resistant)
-> compensatory renin hypersecretion (feedback, no downstream effect)
-> skull ossification defect (large fontanelles, wide sutures; perfusion-linked, inferred)
 - ARM C (hematologic – manifest mainly in survivors):
 - 5c. Loss of angiotensin II pro-erythropoietic action (EPO secretagogue + direct AT1 growth factor)
-> impaired erythropoiesis
-> anemia disproportionate to renal failure

Upstream vs downstream

The **upstream** node is unambiguous: ACE LoF → angiotensin II deficiency. Everything else is **downstream** and convergent. The human genetic evidence (gene-independent severity across *AGT/REN/ACE/AGTR1*; PMID: 22095942) and the mouse genetic evidence (phenotypic equivalence of *Ace*–/–, *Agt*–/–, and *Agtr1a/1b*–/–; PMID: 9860997) jointly establish that the pathology is a readout of the final common endpoint — absent angiotensin II / AT1 signaling and renal hypoperfusion — not of any ACE-specific moonlighting function.

Cell types, tissues, and processes

Level	Structure / process	Ontology suggestion
Organ (primary)	Kidney	UBERON:0002113
Organ (secondary)	Lung (hypoplasia via oligohydramnios)	UBERON:0002048
Organ (secondary)	Skull / cranial bones (ossification defect)	UBERON:0003129
Organ (rare assoc.)	Colon (microcolon; novel RTD assoc., P 30071301)	UBERON:0001155
Tissue	Renal tubular epithelium	UBERON:0004135
Cell	Proximal tubule epithelial cell	CL:1000838
Cell	Juxtaglomerular (renin-producing) cell	CL:1000697
Cell	Erythroid progenitor cell	CL:0000038
Process	Proximal tubule / nephron differentiation	GO:0072014 / GO:0072006
Process	Ureteric bud branching morphogenesis	GO:0060676
Process	Renin–angiotensin regulation of blood pressure	GO:0002018
Process	Regulation of erythropoiesis	GO:0045646
Pathway	Renin–angiotensin system	KEGG hsa04614
Chemical	Angiotensin II	CHEBI:2719

Evidence Base

PMID	Study (abbrev.)	Evidence type	Supports
16116425	Mutations in RAS genes cause AR-RTD (Gribouval 2005)	Human genetic	F1, F2 — ACE as one of 4 RAS genes; hypoperfusion mechanism
16790508	RTD, role of RAS (Lacoste 2006)	Human histopathology	F2, F3 — proximal-tubule defect from 18 wk; renin ↑ ; perinatal death
22095942	Spectrum of RAS mutations in AR-RTD (Gribouval 2012)	Human genetic cohort	F5 — ACE = 64.6% of families; gene-independent severity
32198635	Biallelic RAS mutations can present as progressive CKD (Fila 2020)	Human clinical	F1, F3, F6 — atypical CKD; disproportionate anemia; polyuro-polydipsia
34957720	RTD secondary to RAS mutations (Schreiber 2021)	Human clinical	F3 — survivors 5–20 yr with variable CKD
9860997	AT1A/AT1B double-KO mice	Mouse model	F4 — phenotype "virtually identical" to Ace ^{-/-} , Agt ^{-/-} —
18406804	Role of ACE (review)	Mouse model	F4 — Ace ^{-/-} low BP + renal developmental defects
11096065	Lethality in Agt ^{-/-} mice	Mouse model	F4 — systemic AGT loss → perinatal death, renal lesions
18158355	ACE C-domain main cleavage site	Mouse model	F4 — urinary concentrating defect
26107632	Erythropoiesis/BP via AT1 (Kato 2015)	Mouse model	F6 — Ang II deficiency → anemia, rescued by Ang II
20400218	RAS in regulation of erythropoiesis	Review	F6 — Ang II is a physiological erythropoiesis regulator

PMID	Study (abbrev.)	Evidence type	Supports
24061643 / 21359618	RAS in ureteric-bud branching	Developmental	F5 — Ang II loss impairs UB branching → medullary hypoplasia
30071301	RTD + microcolon	Human clinical	F1 — ACE variants; novel gut association
30058238 / 32329243	Novel ACE splice/compound-het variants	Human genetic	F1 — variant spectrum (splice, frameshift, missense)
33768328	Adult AGTR1 LoF survivor	Human clinical	F3 — survival with managed tubular/glomerular dysfunction
35848000 / 42831903 / 40930738 / 40709124	ACE-RTD neonatal survivors	Human clinical	F3 — survival with dialysis/ventilation

Section-by-Section Characterization

1. Disease Information

RTD-ACE is a severe, prenatally-onset autosomal recessive disorder of kidney development characterized by poor differentiation of renal proximal tubules with a grossly normal kidney architecture, leading to fetal anuria, oligohydramnios, and Potter sequence. **Identifiers:** MONDO:0700337; OMIM 267430 (RTD phenotype); gene *ACE* OMIM 106180; Orphanet ORPHA:97369; ICD-10 Q63.8 (other specified congenital malformations of kidney). **Synonyms:** autosomal recessive renal tubular dysgenesis (AR-RTD); RTD; primary/hereditary RTD; RTD-ACE (ACE-related RTD). The knowledge base is derived from aggregated disease-level resources (OMIM, Orphanet) plus individual-patient case reports and small case series.

2. Etiology

Primary cause: biallelic (homozygous or compound heterozygous) loss-of-function variants in *ACE*. **Genetic risk factors:** consanguinity markedly increases risk (many reported families are consanguineous, e.g., [PMID: 30058238](#), [PMID: 38649831](#)); carrier parents are typically asymptomatic. **Environmental phenocopy:** acquired RTD occurs with in-utero exposure to RAS-blocking drugs (ACE inhibitors, ARBs) and NSAIDs ([PMID: 35848000](#)), and with any cause of severe fetal renal hypoperfusion (e.g., twin-twin transfusion), demonstrating a true **gene–environment convergence** on the same angiotensin II–deficiency / hypoperfusion endpoint ([PMID: 22095942](#)). **Protective factors:** none genetically established; postnatally, intensive supportive care is "protective" against early death.

3. Phenotypes

Phenotype	Type	HPO term	Onset	Frequency
Oligohydramnios/ anhydramnios	Prenatal sign	HP:0001562	Prenatal	Near- universal
Fetal/neonatal anuria/oliguria	Clinical sign	HP:0100519 / HP:0100520	Fetal/ neonatal	Near- universal
Refractory arterial hypotension	Clinical sign	HP:0002615	Neonatal	Very frequent
Pulmonary hypoplasia	Physical	HP:0002089	Prenatal/ neonatal	Very frequent
Large fontanelles / delayed skull ossification	Physical	HP:0000239 / HP:0010656	Neonatal	Frequent
Renal tubular dysgenesis (proximal tubule paucity)	Lab/path	HP:0000107 context	Prenatal	Defining
Chronic kidney disease / ESKD (survivors)	Lab	HP:0012622 / HP:0003774	Childhood	Survivors
Anemia (disproportionate)	Lab	HP:0001903	Childhood	Survivors (atypical)
Polyuria–polydipsia / concentrating defect	Clinical	HP:0000103 / HP:0001959	Childhood	Survivors (atypical)

Severity is classically **severe/lethal**; the course in survivors is **progressive** toward CKD/ESKD. Quality-of-life impact in survivors is substantial (dialysis dependence, growth impairment, transplantation need).

4. Genetic/Molecular Information

Causal gene: *ACE* (gene OMIM 106180; HGNC:2707; 17q23.3). **Variant types:** missense, nonsense, frameshift, canonical splice-site; all loss-of-function. **Examples:** c.1454_1455insC p.(S486Ffs29); c.1037C>G p.(Ser346Trp); c.1465A>G p.(Asn489Asp) ([PMID: 32329243](#)); c.2642-1G>A splice ([PMID: 30058238](#)). **Origin:** germline, biallelic. **Functional consequence:** loss of *ACE* enzymatic activity → no angiotensin II. **Allele frequency:** pathogenic alleles are rare/absent in gnomAD. **Modifier genes:** the other *RAS* genes act as a functional network; no formal modifier established, though gene-independent severity ([PMID: 22095942](#)) argues the pathway endpoint dominates. **Epigenetic/chromosomal:*** no recurrent abnormalities reported.

5. Environmental Information

Non-genetic phenocopies: maternal ACE inhibitors/ARBs and NSAIDs in pregnancy; severe fetal renal hypoperfusion of any cause ([PMID: 35848000](#), [PMID: 22095942](#)). No infectious agents implicated. Relevant chemical entities: captopril/enalapril (ACE inhibitors), NSAIDs.

6. Mechanism / Pathophysiology

See the ordered causal chain above. Core pathway: **renin–angiotensin system (KEGG hsa04614)**. Upstream lesion = *ACE* LoF → angiotensin II deficiency; downstream = failed proximal-tubule differentiation and ureteric-bud branching (renal), loss of vasopressor tone (hemodynamic), and impaired erythropoiesis (hematologic). Compensatory renin hypersecretion is a diagnostic feedback signature ([PMID: 16790508](#), [PMID: 7994002](#)).

7. Anatomical Structures Affected

Primary: kidney (UBERON:0002113), specifically renal cortex proximal tubules and preglomerular vasculature. Secondary: lung (pulmonary hypoplasia, UBERON:0002048), cranial bones (UBERON:0003129), and rarely colon (microcolon, [PMID: 30071301](#)). Body systems: urinary, cardiovascular (hemodynamic), respiratory, skeletal, and hematopoietic. Lateralization: **bilateral** renal involvement.

8. Temporal Development

Onset is **congenital/prenatal** (proximal-tubule defect from ~18 gestational weeks; [PMID: 16790508](#)). Course: historically lethal in the perinatal period; in survivors, **chronic and progressive** toward ESKD over years ([PMID: 34957720](#)). Critical intervention window is the immediate neonatal period (BP support, ventilation, dialysis); prenatal diagnosis enables counseling and planning.

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** high/complete for biallelic LoF, though with variable expressivity (lethal vs survivor-CKD vs atypical late CKD). **Consanguinity** is a major contributor. **Carrier frequency:** low; founder effects not firmly established. **Epidemiology:** very rare (Orphanet class <1/1,000,000); exact prevalence/incidence not well quantified. **Sex ratio:** approximately equal (autosomal). **ACE** accounts for ≈65% of genetically solved families ([PMID: 22095942](#)).

10. Diagnostics

Prenatal: oligohydramnios/anhydramnios on ultrasound; enlarged echogenic kidneys with poor corticomedullary differentiation. **Postnatal labs:** anuria, refractory hypotension, markedly elevated plasma renin activity with undetectable/low ACE ([PMID: 32808512](#)). **Imaging:** skull X-ray showing large fontanelles/wide sutures; renal ultrasound. **Histopathology (definitive morphology):** absent/paucal differentiated proximal tubules with increased renin immunostaining. **Genetic testing (definitive):** WES/WGS or RAS-gene panels (*ACE*, *AGT*, *REN*, *AGTR1*) — WES identified ACE variants in multiple cases ([PMID: 30058238](#), [PMID: 35848000](#)). **Differential diagnosis:** ARPKD, bilateral renal agenesis/dysplasia, other CAKUT, and acquired (drug-induced) RTD. Prenatal diagnosis is available for at-risk pregnancies after family variant identification.

11. Outcome/Prognosis

Historically very poor — near-universal perinatal death ([PMID: 16790508](#)). With modern intensive care and dialysis, survivors reach 5–20 years with variable CKD/ESKD ([PMID: 34957720](#)); an adult AGTR1 survivor retained GFR ~30 mL/min with managed tubular/glomerular dysfunction ([PMID: 33768328](#)). Complications: pulmonary hypoplasia, ESKD, growth impairment, disproportionate anemia. Prognostic factors: degree of pulmonary hypoplasia, postnatal urine-output trajectory, and feasibility of dialysis.

12. Treatment

No curative therapy; management is **supportive**. Neonatal: mechanical ventilation, aggressive vasopressor/blood-pressure support, early peritoneal dialysis (PMID: 42831903, PMID: 40930738). Longer-term: renal replacement therapy and eventual kidney transplantation; fludrocortisone has been used to treat hyperkalemia/salt-wasting in a RAS-LoF survivor (PMID: 33768328); erythropoiesis-stimulating agents for the anemia. RAS-blocking drugs are **contraindicated** (would worsen the deficiency). No gene/cell/RNA therapies exist. NCIT terms: Dialysis (NCIT:C15248), Kidney Transplantation (NCIT:C15366), Supportive Care (NCIT:C15285).

13. Prevention

No primary prevention of the genetic form. **Secondary/tertiary**: avoid RAS inhibitors and NSAIDs in pregnancy (prevents acquired phenocopy); prenatal and preimplantation genetic diagnosis for at-risk couples after variant identification; **genetic counseling** for consanguineous and previously affected families; cascade carrier testing. Early prenatal detection enables planned delivery at a tertiary center with neonatal dialysis capability.

14. Other Species / Natural Disease

No naturally occurring companion-animal RTD-ACE is well documented; the disease is primarily defined in humans. Orthologous genes are conserved — mouse *Ace* (NCBI Gene 11421), *Agt*, *Ren*, *Agtr1a/Agtr1b*. The renal mechanism is **evolutionarily conserved** across mammals, as evidenced by matching mouse knockout phenotypes. Experimental species: *Mus musculus* (NCBI:txid10090), *Rattus norvegicus*.

15. Model Organisms

The mouse is the primary model. Relevant genetic models: *Ace*^{-/-} (low BP, renal developmental defects; PMID: 18406804); *Agt*^{-/-} (perinatal lethal, renal vascular/tubular lesions; PMID: 11096065); *Agtr1a*^{-/-}*Agtr1b*^{-/-} double knockout (renal vascular thickening, medullary atrophy; PMID: 9860997); ACE C-domain-inactive knock-in (urinary concentrating defect; PMID: 18158355). **Phenotype recapitulation**: high — these models reproduce the renal vascular/tubular pathology, hypotension, concentrating defect, and anemia. **Limitations**: mouse models capture the renal and hemodynamic lesions but the human Potter-sequence/pulmonary-hypoplasia context differs, and the atypical human progressive-CKD phenotype is less directly modeled.

Limitations and Knowledge Gaps

1. **Rarity and small cohorts.** Most evidence derives from single case reports and modest series; precise prevalence, incidence, penetrance, and survival statistics are not firmly established.
2. **Genotype–phenotype resolution.** Although severity appears gene-independent, it remains unclear what determines whether a given biallelic *ACE* LoF produces lethal neonatal disease versus the atypical progressive-CKD phenotype — residual ACE activity, modifier genes, or environmental factors are all plausible but untested.
3. **Inferred steps in the causal chain.** The link from angiotensin II deficiency to skull ossification defects, and the ureteric-bud branching contribution in humans, is extrapolated largely from mouse/developmental data rather than demonstrated directly in human RTD-ACE tissue.
4. **Survivor natural history.** Long-term outcomes (renal, neurocognitive, cardiovascular) in the growing survivor population are incompletely characterized; the oldest well-documented survivors are only ~20 years old.
5. **No disease-specific therapeutics.** No treatment restores fetal angiotensin II signaling during the critical developmental window; management is entirely supportive.

Proposed Follow-up Experiments / Actions

1. **International RTD-ACE registry** to quantify prevalence, survival with modern care, and long-term survivor outcomes, enabling evidence-based prognostic counseling.
2. **Systematic ACE genotype–residual-activity correlation** (functional assays of patient variants) to test whether partial/hypomorphic activity predicts the atypical progressive-CKD phenotype versus lethal disease.
3. **Kidney-organoid and iPSC models** carrying patient *ACE* variants to dissect the human proximal-tubule differentiation defect and test whether exogenous angiotensin II (or AT1 agonism) during a developmental window rescues tubulogenesis.
4. **Prospective anemia characterization in survivors** — measure EPO, reticulocyte response, and ESA responsiveness to confirm the angiotensin II–erythropoiesis mechanism clinically and optimize anemia management.
5. **Standardized neonatal management protocol** (BP targets, timing of dialysis, ventilation strategy) evaluated prospectively, given the expanding survivor spectrum.

6. Pharmacovigilance / public-health messaging reinforcing avoidance of RAS inhibitors and NSAIDs in pregnancy to prevent acquired phenocopies.

Report compiled from 6 confirmed findings and 42 reviewed papers over 5 investigation iterations. Evidence types are distinguished throughout as human clinical, human genetic/histopathology, mouse model, or review.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery